

SensorHub Rev 2.1 Design Specification

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1. Overview

The SensorHub Rev 2.1 is a compact multi-sensor Bluetooth Low Energy module designed for environmental monitoring and motion detection. It integrates an ESP32-S3 MCU with temperature, humidity, and accelerometer sensors in a 32x24mm form factor.

2. Electrical Specifications

Supply Voltage: 3.0V to 5.5V via USB-C or LiPo battery
Regulated Output: 3.3V +/- 2% from TPS63020 buck-boost
Active Current: 356mA peak during BLE TX at +9dBm
Sleep Current: less than 12uA in deep sleep with RTC
Operating Temperature: -20C to +60C
Storage Temperature: -40C to +85C

3. Pin Assignment

GPIO8 - I2C SDA with 4.7k pull-up to 3.3V
GPIO9 - I2C SCL with 4.7k pull-up to 3.3V
GPIO10 - SPI CS active low
GPIO11 - SPI MOSI
GPIO12 - SPI MISO
GPIO13 - SPI CLK
GPIO43 - UART TX debug 115200 baud
GPIO44 - UART RX debug
GPIO1 - ADC VBAT monitor 100k/100k divider
GPIO4 - Status LED active high
GPIO5 - User button active low with internal pull-up
GPIO6 - LIS3DH INT1 motion interrupt

4. I2C Bus Configuration

Bus Speed: 400kHz Fast Mode
SHT40 Address: 0x44 temperature and humidity sensor
LIS3DH Address: 0x19 three-axis accelerometer
AT24C256 Address: 0x50 32KB EEPROM for config storage

5. Power Architecture

The power subsystem uses a TPS63020 buck-boost converter to produce a stable 3.3V rail from either USB 5V or single-cell LiPo battery. Battery charging is handled by the MCP73831 linear charger IC.

Power Budget:

ESP32-S3 active BLE TX:	350mA
SHT40 measurement:	0.4mA
LIS3DH normal mode:	0.5mA
AT24C256 write:	5.0mA
Misc LED and pull-ups:	1.0mA
Total active:	~357mA

6. BLE Configuration

Protocol: Bluetooth 5.0 Low Energy
GATT Service: Environmental Sensing UUID 0x181A
Advertising Interval: 1000ms normal or 200ms fast connect
TX Power: 0dBm default and +9dBm maximum
Bonding: Yes with up to 4 paired devices
Encryption: AES-128-CCM

7. Mechanical

PCB Dimensions: 32.0 x 24.0 x 1.0mm
Layer Count: 4 layers SIG/GND/PWR/SIG
Surface Finish: ENIG
Mounting: 4x M2 holes at corners
Weight: approximately 6g excluding battery

8. Compliance

FCC Part 15 certification pending
CE RED 2014/53/EU certification pending
RoHS 3 compliant
REACH SVHC declaration available on request

9. Revision History

Rev A	2026-05-10	Initial release
Rev B	TBD	Add thermal management section